Experiment 4

OLAP Operations (Slice, Dice, Drill-up & Drill-down) using Microsoft Power BI

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Aim

To perform Online Analytical Processing (OLAP) operations using Microsoft Power BI by creating multidimensional reports and applying Slice, Dice, Drill-down, and Drill-up operations on the Global Superstore dataset for analytical decision-making.

Tools / Software Required

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor (Built into Power BI Desktop)
4. DAX (Data Analysis Expressions)

Theory

Online Analytical Processing (OLAP) is a Business Intelligence technique used for multidimensional analysis of business data. OLAP enables users to analyze information from different perspectives and perform complex analytical queries efficiently.

Common OLAP operations include:

Slice: Selecting a single dimension value from a dataset.
Dice: Selecting multiple dimensions and values simultaneously.
Drill-down: Moving from summarized data to detailed data.
Drill-up: Moving from detailed data to summarized data.

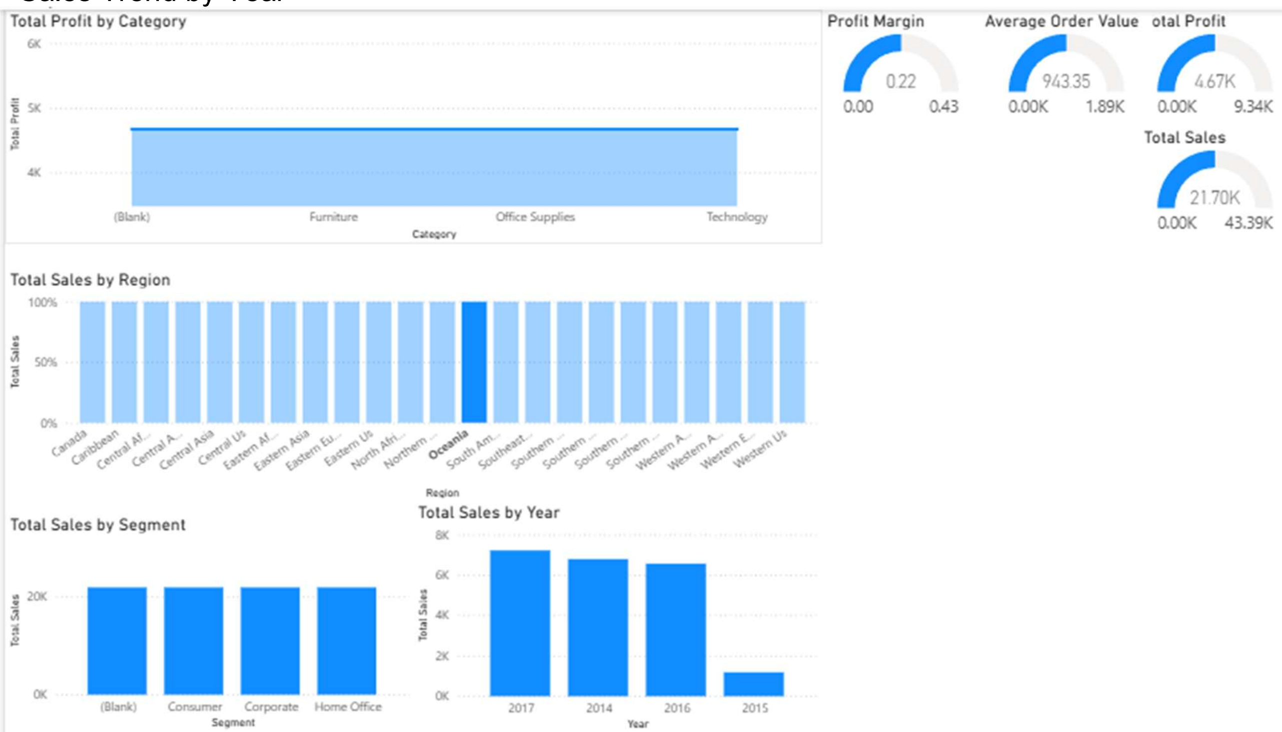
Power BI supports OLAP-style analysis through hierarchies, filters, slicers, visual interactions, and drill functionality.

Conduct

Conduct - Part A: Load Dataset and Build Report

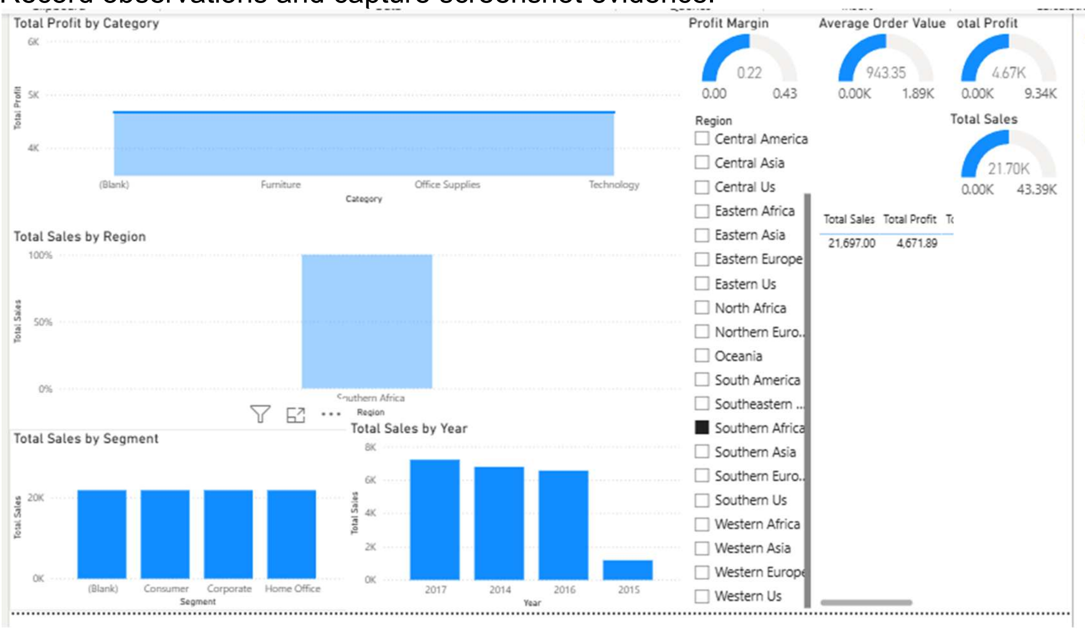
Step 1: Open Power BI Desktop.
Step 2: Open the Power BI model created in Experiment 3 or import the Global Superstore dataset.
Step 3: Verify relationships between Orders, Returns, and People tables.
Step 4: Create report visuals:

• Sales by Region
• Profit by Category
• Sales by Segment
• Sales Trend by Year



Conduct - Part B: Slice Operation

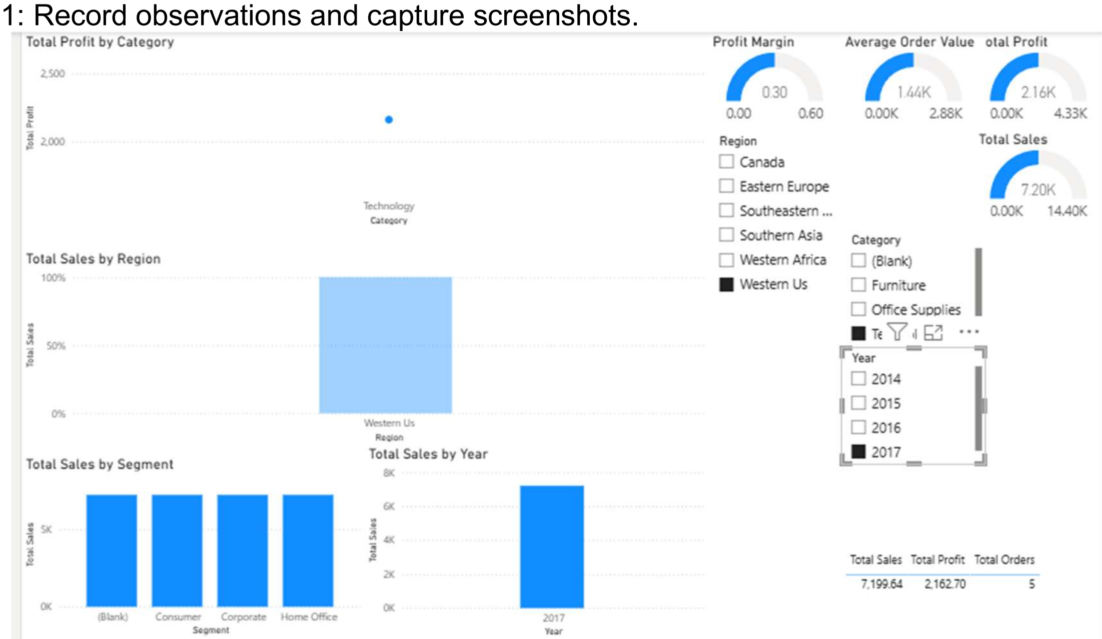
Step 5: Add a slicer visual using Region.
Step 6: Select a single Region such as West.
Step 7: Observe changes in Sales, Profit, and Orders.
Step 8: Record observations and capture screenshot evidence.



Conduct - Part C: Dice Operation

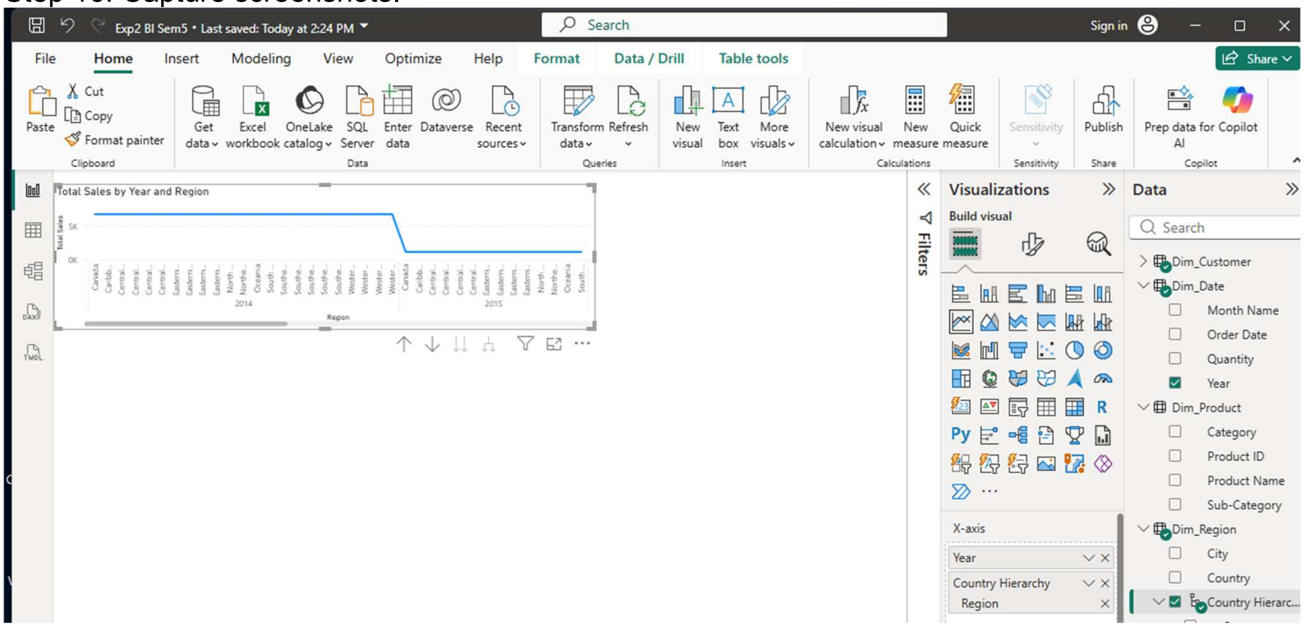
Step 9: Apply multiple filters simultaneously.
Example:
• Region = West
• Category = Technology
• Year = 2022

Step 10: Analyze filtered business metrics.
Step 11: Record observations and capture screenshots.



Conduct - Part D: Drill-down Operation

Step 12: Create a hierarchy using Region → State → City.
Step 13: Create a bar chart using the hierarchy.
Step 14: Use Drill-down mode to navigate from Region level to City level.
Step 15: Observe detailed sales patterns.
Step 16: Capture screenshots.



Conduct - Part E: Drill-up Operation

Step 17: Navigate back from City level to State level and then Region level.
Step 18: Compare aggregated values at each level.
Step 19: Record observations.
Step 20: Capture screenshots.

Conduct - Part F: OLAP Analysis

Step 21: Compare results obtained from Slice, Dice, Drill-down, and Drill-up operations.
Step 22: Create a summary table documenting observations.
Step 23: Explain how each operation supports business decision-making.

Result

OLAP operations including Slice, Dice, Drill-down, and Drill-up were successfully performed using Microsoft Power BI. Multidimensional analysis was carried out on the Global Superstore dataset, enabling detailed and summarized views of business performance from multiple perspectives.

Learning Outcomes

• Explain OLAP concepts and multidimensional analysis. (CO3-BT3)
• Perform Slice and Dice operations using filters and slicers. (CO3-BT3)
• Create hierarchies and perform Drill-down and Drill-up analysis. (CO5-BT6)
• Interpret analytical insights generated through OLAP operations. (CO3-BT4)
• Design interactive analytical reports for business users. (CO5-BT6)

Text / Reference Books

1. Sharda, Delen & Turban, Business Intelligence and Analytics: Systems for Decision Support.
2. Cindi Howson, Successful Business Intelligence.
3. Microsoft Power BI Official Documentation.
4. Microsoft Power Query Documentation.